TEXORA · LECTURE SHEET

Tools of Planning

The planner's instrument set in a garment factory: the T&A calendar, capacity and load planning, line plans and operation bulletins, Gantt charts and CPM networks, cut plans and the daily control reports that keep the plan honest.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 4 · LEVEL 4 TERM 2

1. What planning must answer

Planning converts the order book into dated, resourced commitments (Topic 1). For every order it must answer: **can we** (capacity vs load), **when exactly** (the calendar), **where** (which line, which system), **with what** (materials, machines, skills) and **how will we know** (control reports). Each tool below answers one of those questions; together they form the planning system. Inputs are IE data (SMV, Topic 3), line capacities, material lead times and the shipment dates; outputs are commitments the whole factory can be managed against.

2. The core tools

TOOL	QUESTION IT ANSWERS	FORM / KEY FEATURES
T&A calendar	When exactly — per order.	Dated milestone list per order (approvals, materials, cutting, sewing, shipment); reviewed weekly against actuals.
Capacity plan	Can we — factory level.	Available minutes per line per period (operators × minutes × planned efficiency) vs minutes demanded by confirmed + expected orders; exposes over/under-booking months ahead.
Line plan (loading plan)	Where and when — per line.	Each line's styles sequenced with start/finish dates, quantities, learning-curve

		allowances and changeover gaps.
Operation bulletin	How the garment will be made.	Operation list with SMV, machine, attachments, section targets — the line's build sheet, from IE.
Gantt chart	When — visually, across parallel work.	Bars on a time axis per activity/line; overlaps, float and today's position read at a glance (§3).
CPM / PERT network	What is critical.	Activity network with dependencies; the longest path (critical path) sets the shortest possible duration; float shows where slips are free (§4).
Cut plan	With what — fabric conversion.	Marker ratios, lays and fabric allocation per order (Topic 5); feeds the sewing plan its bundles.
Control reports	How will we know.	Daily production vs target, WIP report, efficiency report, T&A exception list — the plan's feedback loop.

3. Gantt chart

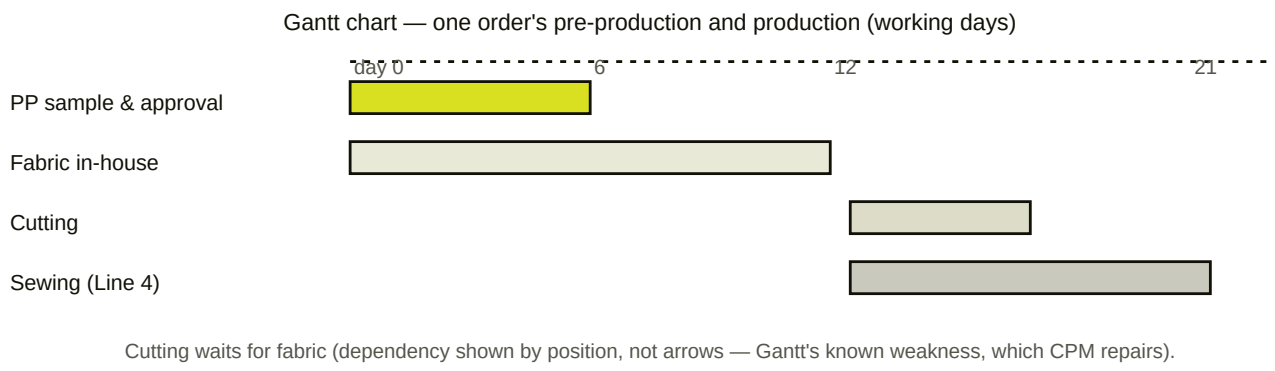


Figure 1 — A planning Gantt: bar positions encode both schedule and dependency. Bar lengths here are 6, 12, 4.5 and 9 days by the day scale above (0–21+ days).

4. CPM: finding the path that decides the date

WORKED CPM — PRE-PRODUCTION NETWORK, DURATIONS IN DAYS

Activities: **A** pattern & sample (3) → then in parallel **B** fabric approval route (2) and **C** trims & accessories route (4) → then **D** bulk cutting release (2) after both. Paths: A–B–D = 3 + 2 + 2 = **7 days**; A–C–D = 3 + 4 + 2 = **9 days**. The **critical path is A–C–D (9 days)**: trims, not fabric, decide this order's earliest cutting date. B carries **2 days of float** (9 − 7) — a 2-day fabric slip is free; a 1-day trims slip moves the shipment plan by a day. Planning effort, expediting and management attention belong on the critical path; that allocation of attention is the entire practical value of the method.

5. Planning discipline

- **One plan, one owner:** the master plan lives in one place (increasingly an ERP/planning board); departmental side-plans that disagree with it are fiction generators.
- **Freeze windows:** the next days are frozen (no re-sequencing except emergencies), the next weeks are firm, beyond that flexible — change is priced by horizon.
- **Plan at realistic efficiency:** loading lines at 100% of the master equation guarantees lateness; plan at demonstrated efficiency with the learning curve for new styles (Topic 1).
- **Review rhythm:** daily output vs plan on the floor, weekly T&A and capacity review in management — slippage found weekly is recoverable, found monthly is freight.

Glossary

TERM	MEANING
Load	Minutes of work demanded of a line/factory in a period, from orders × SMV.
Critical path	The longest dependency chain in a network; its length is the shortest possible project duration.
Float	Time an activity can slip without delaying the project end (total float).
Operation bulletin	The garment's build sheet: operations, SMVs, machines, attachments and section targets.
Freeze window	Near-term period in which the plan may not be re-sequenced except by escalation.

Key points and summary

- Tools divide the planning questions: T&A (when), capacity/load (can we), line plan (where), operation bulletin (how), Gantt/CPM (what is critical), control reports (how we know).
- Gantt shows schedule; CPM adds dependency logic and identifies where slippage is free (float) and where it is fatal (critical path).
- Plans are kept honest by freeze windows, realistic efficiency loading and a fixed review rhythm.

Practice

Answer in your own words; brief model answers follow.

1. Network: P (2 days) → Q (3) and R (5) in parallel → S (1). Find both path lengths, the critical path and Q's float.
2. A line has 30 operators at 55% planned efficiency. How many standard minutes per day can the capacity plan honestly load onto it?
3. Why does a Gantt chart alone mislead a planner about dependencies?
4. An order's fabric is 3 days late but the T&A shows shipment still protected. Using CPM language, explain how that can be true — and when it stops being true.
5. What two IE documents must exist before a line plan for a new style can be more than a guess?

Answer key

1. P–Q–S = 2 + 3 + 1 = 6 days; P–R–S = 2 + 5 + 1 = 8 days. Critical path P–R–S (8 days). Q's float = 8 − 6 = 2 days.
2. Attended = 30 × 480 = 14,400 min; loadable standard minutes = 14,400 × 0.55 = 7,920 min/day.
3. Bars show when activities run but not why: two bars can sit side by side with a hard dependency, a soft one, or none. Re-sequencing from the picture alone can break a hidden dependency; the network diagram carries the logic explicitly.
4. The fabric route is not on the critical path and its float (≥ 3 days) absorbs the slip. It stops being true when cumulative slip exceeds the float — at that moment the fabric route becomes critical and every further day moves the shipment plan.
5. The operation bulletin (operation SMVs and machines) and the garment SMV/capacity data behind it — without times, "loading" a line with styles is counting garments, not minutes, and unlike styles make the count meaningless.